Higher National Diploma in Accountancy
Second Year, Second Semester Examination – 2020
HNDA- 2201 Cost and Management Accounting

Instructions for candidates:

No of Questions: 06

Answer any 05 questions

No of pages: 07

All the questions carry equal marks

Time: **03 hours**

Question 01

I. List out four differences between **cost and management Accounting** and **Financial Accounting.** **(03 Marks)**

II. Define the following terms with examples.

- Cost unit
- Cost Centre
- Variable cost
- Fixed cost **(08 Marks)**

III. State which method of costing you would recommend for use in the following industries.

a. Hospitals
b. Building construction
c. Cement manufacturing
d. Shoe manufacturing
e. Biscuit manufacturing
f. Furniture manufacturing **(03 Marks)**

IV. Company XYZ Ltd which is engaging a shoe manufacturing. According to the production manager, the production information is available for March 2020 is as follows:

Raw material cost per unit is Rs. 25
Total number of shoe manufacture is 1,000
Labour charge is Rs.35 per hour
Time taken to produce a shoe is 30 minutes

The total cost of production is Rs. 60,000
Calculate the Fixed Cost of production for XYZ Ltd in March 2020. **(03 Marks)**

V. PR Company manufactures a single product, M. Budgeted production output of product M during August is 200 units. Each unit of product M requires 6 labour hours for completion and PR Co. anticipates 20 per cent idle time. Labour is paid at a rate of Rs.7 per hour. What is the direct labour cost for August?

(03 Marks)

(Total Marks 20)

Question 02

(A) The Gee Company uses about 75,000 items per year and the usage is fairly constant at 6,250 per month. The items cost Rs. 1.50 per unit when purchase in quantities and carrying cost is estimated to be 20% of average inventory investment on the annual basis. The cost to place the order and to process the delivery is Rs. 18.

It takes 45 days to receive delivery from the date of an order and a safety stock of 3,250 units is desired.

You are required to determine

I. Economic Order Quantity (EOQ) and frequency of orders. **(04 Marks)**
II. Re Order Level. **(02 Marks)**

(B) P.Ltd is a manufacturing company having three production departments A, B and C, and two service departments X and Y. The following is the budget for December, 2021:

	Totals Rs.
Rent and rates	5,000
General lighting	600
Indirect wages	1,500
Power	1,500
Depreciation of machinery	10,000
Sundry expenses	1,000

dditional information:

	A	B	C	D	E
ea (sq. m)	2,000	2,500	3,000	2,000	500

Value of machinery(Rs)	60,000	80,000	100,000	5,000	5,000
Machine hour	1,000	2,000	4,000	1,000	1,000
Light points(nos)	10	15	20	10	5
Direct wages	3,000	2,000	3,000	1,500	500
H.P. of machines	60	30	50	10	-
Working hours	6,226	4,026	4,066	-	-

The expenses of D and E are allocated as follows;

D	20%	30%	40%	-	10%
E	40%	20%	30%	10%	-

You are required to do the following:

i) A statement showing distribution of overheads
ii) A statement showing re-distribution of service department cost to production department
iii) Compute overhead absorption rate
iv) Ascertain total cost of an article if material cost is Rs 50, labour cost is Rs 30 and it passes through departments A,B and C for 4,5 and 3 hours respectively. **(14 Marks)**

(Total 20 Marks)

Question 03

(A) State two advantages of budgetary system. **(02 Marks)**

(B) Briefly explain the concept of **"flexible budget"** **(02 Marks)**

(C) Sujith Ltd, a manufacturing company provides you the following data for next budgeted period. Using the following data, prepare a cash budget for the month of July, August and September of 2020.

Month	Total sales(Rs)	Purchases(Rs)	Wages(Rs)	Production. OH(Rs)	Selling OH(Rs)	Administrative OH(Rs)
May	240,000	120,000	20,000	16,000	23,000	15,000
June	260,000	100,000	24,000	20,000	24,000	16,000
July	160,000	108,000	16,000	18,000	18,000	12,000
Aug	232,000	112,000	20,000	17,000	22,000	14,000
Sep	176,000	90,000	16,000	14,000	16,000	15,000

Additional information

i. Cash balance on 01/07 is Rs. 30,000.
ii. Assume cash sales to be 50% of total sales.
iii. Credit sales are collected in the following manner.
   20% realized in the month of sales, discount allowed 2%.
   Balance realize equally in two subsequent months.
iv. Sales commission at 4% on total sales is to be paid within the month following active sales.
v. Payments to creditors- 60% will be settled in the same month, remaining 40% in the following month.(Assume that all purchases will be on credit basis)
vi. All Overhead is paid with one month arrears and administrative OH includes monthly depreciation of Rs. 6,000.
vii. Wages are paid instantly.
viii. A bank loan of Rs. 50,000 will be taken during the month of August.
ix. A new machine is to be installed at Rs. 30,000 on credit during the month of July and the consideration has to be paid in 3 equal installments during the month of August, September and October.
x. Income tax of Rs. 60,000 to be paid on 15/09. **(16 Marks)**

(Total 20 Marks)

Question 04

(A) Define the term "breakeven point" **(02 Marks)**

(B) Nimsara Company plans to sell 120,000 units of product A in a period at a selling price of Rs. 15 per unit. Fixed overhead and net profit for the period are expected to be Rs. 440,000 and Rs. 520,000 respectively using the existing production process.
The company is considering a change to its production process. The change would increase the fixed overheads to Rs. 700,000 in the period and reduce the variable cost to Rs. 5 per unit. The selling price will remain constant regardless of production process. Production capacity in both the existing and changed processes would be 160,000 units in the period.

I. Required to calculate followings for the existing production process
   (a) Breakeven point **(03 Marks)**
   (b) Margin of safety as a percentage of sales **(02 Marks)**
   (c) Contribution sales ratio **(01 Mark)**

II. Advice management, using supporting calculations, whether to change the production process if sales are 120,000 units in the period. **(04 Marks)**

III. Assume that under the change of production process, firm can increase its sales units by 40,000 units, by reducing the selling price per unit by Rs.1 and extra advertising cost of Rs. 50,000. Calculate expected net profit. **(02 Marks)**

(C) Dilmith ltd is manufacturing three products X, Y and Z and selling them in a competitive market. Details of current demand, selling price and cost structure are given bellow.

Products	X	Y	Z
Expected Demand (units)	20,000	24,000	40,000
Selling price per unit (Rs)	40	32	20
Variable cost per unit			
Direct material Rs. 20 per Kg	12	8	4
Direct labour Rs. 3 per hour	6	6	3
Variable overhead	4	2	2

Fixed overhead cost per period Rs. 276,000.

The company is frequently affected by acute scarcity of raw material and high labour turnover. During the next period, it is expected that raw material available will be only 24200 Kg.

You are required to prepare best production plan and determine the expected profit.

(06 Marks)

(Total 20 Marks)

Question 05

(A) Define "**job costing**" with its special features. **(03 Marks)**

(B) A factory uses job costing system. The following data obtained on job no.400 for the year 2019/2020.

Direct material Rs. 40,000

Wages

Department A- 60 hours at Rs. 30 per hour

B – 40 hours at Rs. 20 per hour

C – 20 hours at Rs.50 per hour

Variable overhead expenses for three departments were estimated as follows.

Department A Rs. 50,000 for 5,000 labour hours

B Rs. 30,000 for 1,500 labour hours

C Rs. 20,000 for 500 labour hours

Fixed overhead: Estimated at Rs. 200,000 for 10,000 normal working hours.

You are required to calculate the cost of Job no. 400 and calculate the price to give by adding profit of 25% on job cost. **(05 Marks)**

(C) A product passes through two process A and B. The following data relates to the process "A" for the month of January.
1,800 units were in process at the begging value at Rs. 83,200 made up of Rs.38, 400 of material cost, Rs. 24,000 of labour cost and Rs. 20,800 of overhead expenditure.

12,000 units of material were introduced into the process during the period and cost of these units were Rs. 211,680. Direct wages and overhead expenditure incurred this period were Rs. 272,400 and 132,080 respectively.
There were 2,400 units in process at the end of the period. These units were 75% complete with regard to direct material, 50% complete with respect to labour and 30% complete with overhead cost.
Expected normal loss in process A is 10% of input units.
At the end of the period 9,500 units have been transferred to the "process B."
The units scrap to be considered as 100% completed in all elements.
Units scrap can be sold at Rs. 5.00 per unit.

You are required to prepare followings by using weighted average method.

a) Process A account
b) Statement of production, cost and evaluation
c) Abnormal loss/gain account

(12 Marks)
(20 Marks)

Questions 06

You are given following information relating to a particular product.

<u>Standard cost data per unit</u>

Direct material

A	5 Kg at Rs. 36 per Kg	7580
B	4 Kg atRs. 40 per Kg.	8000

Direct Labour
3 hours at Rs. 50 per hour
Variable overhead 3 hours at Rs. 30 per hour.

Budgeted Fixed overhead for the period Rs. 360,000
Budgeted production 1200 units.

Actual results for the period.

Direct material – A 9,200 Kg at 39 per kg

B 7,000Kg at 42 per Kg

Direct labour 5,000 hours at Rs. 55 per hour.

Variable overhead – 5,000 hours at Rs. 36 per hour.
Fixed overhead Rs. 450,000.
Actual production 1,500 units.

You are required to compute following variances.

I. Direct material cost variance
II. Direct material price variance
III. Direct material usage variance
IV. Direct material mix variance
V. Direct material yield variance
VI. Labour rate variance
VII. Labour efficiency variance
VIII. Variable overhead efficiency variance
IX. Variable overhead expenditure
X. Fixed overhead expenditure variance

(Total 20 Marks)